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VIA ASSEMBLY FOR PRINTED CIRCUIT BOARD

Abstract

Provided for herein is an apparatus that includes a layer of a printed circuit board, a first pair of signal vias extending through the layer, a second pair of signal vias extending through the layer, a first plurality of ground vias extending through the layer, and a second plurality of ground vias extending through the layer. Each of the pairs of signal vias are configured to propagate respective signals. The first plurality of ground vias at least partially circumferentially surround a first signal via of the first pair of signal vias, and the second plurality of ground vias at least partially circumferentially surround a second signal via of the second pair of signal vias to reduce interference of electrical fields emitted by the pairs of signal vias. The first plurality of ground vias and the second plurality of ground vias share a common ground via.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to printed circuit boards (PCBs).

BACKGROUND

[0002] A printed circuit board (PCB) electrically couples various electronic components with one another. For example, a PCB may include multiple layers, each having different electronic components and traces routed along the layers to electrically couple to electronic components of the same layer. Additionally, the PCB may include vias that extend between layers to electrically couple electronic components of different layers to one another. For example, a signal may propagate from a first electronic component of a first layer, through a first trace routed along the first layer, through a via extending from the first layer to a second layer, through a second trace routed along the second layer, and to an electronic component of the second layer. Unfortunately, the PCB may be subject to crosstalk in which transmitted signals interfere with one another. For example, electric fields emitted by vias during signal transmission may overlap with one another. Interference between signals may increase signal loss, thereby reducing signal integrity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a side perspective cross-sectional view of a printed circuit board (PCB) that includes multiple layers, according to an example embodiment.

[0004] FIG. **2** is a side view of a PCB that includes multiple layers, according to an example embodiment.

[0005] FIG. **3** is a top perspective view of the PCB of FIG. **2**.

[0006] FIG. **4** is a schematic diagram of an external layer of a PCB with ground vias surrounding pairs of signal vias, according to an example embodiment.

[0007] FIG. **5** is a schematic diagram of an internal layer of a PCB with ground vias surrounding pairs of signal vias, according to an example embodiment.

[0008] FIG. **6** is a schematic diagram of a power layer of a PCB having a ground portion with ground vias surrounding pairs of signal vias, according to an example embodiment.

[0009] FIG. **7** is a flowchart of a method for manufacturing a PCB with ground vias surrounding pairs of signal vias, according to an example embodiment.

[0010] FIG. **8** is a flowchart of a method for manufacturing a PCB that includes a power layer having a ground portion with ground vias surrounding pairs of signal vias, according to an example embodiment.

DETAILED DESCRIPTION

Overview

[0011] Techniques are provided herein for reducing crosstalk for a printed circuit board (PCB). In some aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; a first pair of signal vias extending through the layer and configured to propagate respective signals; a second pair of signal vias extending through the layer and configured to propagate respective signals; a first plurality of ground vias extending through the layer and at least partially circumferentially surrounding a first signal via of the first pair of signal vias; and a second plurality of ground vias extending through the layer and at least partially circumferentially surrounding a second signal via of the second pair of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias share a common ground via.

[0012] According to other aspects, the techniques described herein relate to a method including:

forming adjacent pairs of signal vias through a layer of a printed circuit board; forming a first plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a first signal via of a first pair of signal vias of the adjacent pairs of signal vias; and forming a second plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a second signal via of a second pair of signal vias of the adjacent pairs of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias comprise a common ground via.

[0013] In still other aspects, the techniques described herein relate to an apparatus including: a first pair of signal vias of a printed circuit board, wherein the first pair of signal vias are configured to propagate respective signals; a second pair of signal vias of the printed circuit board, wherein the second pair of signal vias are configured to propagate respective signals, and the signal vias of the first pair of signal vias and the second pair of signal vias are offset from one another along a first axis and aligned with one another along a second axis that is perpendicular to the first axis; and a single ground via positioned between the first pair of signal vias and the second pair of signal vias along the first axis.

Example Embodiments

[0014] Techniques discussed herein are related to a PCB having ground via arrangements to reduce crosstalk. The PCB may include adjacent pairs of signal vias configured to propagate respective signals. The PCB also may include ground vias that are arranged around each pair of signal vias to block electric field leakage between the pairs of signal vias. For example, a first plurality of ground vias may circumferentially surround a first signal via of one of the pairs of signal vias, and a second plurality of ground vias may circumferentially surround a second signal via of an adjacent pair of signal vias. The first plurality of ground vias and the second plurality of ground vias may share a common ground via positioned between the adjacent pairs of signal vias. For instance, the common ground via may be positioned equidistant to the first signal via and the second signal via.

[0015] Such an arrangement of ground vias and signal vias may enable the pairs of signal vias to be positioned more adjacent to one another while sufficiently blocking interference between electric fields. For instance, a single ground via may be positioned between the adjacent pairs of signal vias and block overlap between respective electric fields emitted by both pairs of signal vias. By positioning the pairs of signal vias more adjacent to one another, the space of the PCB may be more efficiently utilized, such as by increasing a quantity of components (e.g., a quantity of signal vias) positioned thereon.

[0016] With reference made to FIG. **1**, depicted therein is a cross-sectional view of a PCB **100** having multiple layers **102**. Each layer **102** may include different electronic components that are electrically coupled to one another. By way of example, a first layer **102**A (e.g., a top layer) may include multiple pads **104** that are exposed to an exterior environment of the PCB **100**. The pads **104** may enable electrical coupling of the PCB **100** to a separate component, such as an integrated circuit (IC), such as an application-specific IC (ASIC). For example, an interconnect **106**, such as a solder ball, may be used to electrically couple one of the pads **104** to the separate component. A trace **108** may also be electrically coupled to the pad **104** and routed along the first layer **102**A to electrically couple the pad **104**, and therefore the interconnect **106** and the separate component electrically coupled to the pad **104**, to an electronic component (e.g., another IC, a resistor, a transistor, a capacitor, a switch, an inductor, a transformer, a sensor, a diode, a relay) of the first layer **102**A.

[0017] Moreover, the PCB **100** may include signal vias **110** that extend through multiple layers **102** of the PCB **100** to electrically couple electronic components of different layers **102** to one another. For instance, a first signal via **110**A may be electrically coupled to the pad **104** that is electrically coupled to the interconnect **106**, and the first signal via **110**A may extend from the first layer **102**A to a second layer **102**B (e.g., an inner layer, a mid layer) of the PCB **100**. A trace routed along the second layer **102**B may be electrically coupled to the first signal via **110**A and to an electronic component of the second layer **102**B. As such, the first signal via **110**A may help electrically couple the electronic component of the second layer **102**B to the electronic component of the first layer **102**A and/or to the separate component electrically coupled to the pad **104**.

[0018] By way of example, during operation of the PCB **100**, a signal may be transmitted between the electronic component of the first layer **102**A, the electronic component of the second layer **102**B, and/or the separate component electrically coupled to the PCB **100**. For instance, the separate component may transmit the signal to the pad **104** by way of the interconnect **106**, and the signal may propagate through the trace **108** secured/connected to the pad **104** and routed along the first layer **102**A toward the electronic component of the first layer **102**A and/or through the first signal via **110**A toward the second layer **102**B, through the trace routed along the second layer **102**B and electrically coupled to the first signal via **110**A, and toward the electronic component of the second layer **102**B.

[0019] However, an integrity of the signal propagated through the first signal via **110**A may be reduced as a result of electrical field interferences. For example, the first signal via **110**A may extend from the first layer **102**A to a third layer **102**C (e.g., a bottom layer) and may transmit a signal from the first layer **102**A toward the third layer **102**C. A second signal via **110**B may extend from the third layer **102**C and terminate prior to the first layer **102**A and may transmit a signal from the third layer **102**C toward the first layer **102**A. Signal transmission along the first signal via **110**A toward the third layer **102**C may cause an electrical field to be emitted from the first signal via **110**A toward the second signal via **110**B, and signal transmission along the second signal via **110**B from the third layer **102**C toward the first layer **102**A may cause an electrical field to be emitted from the second signal via **110**B toward the first signal via **110**A. Consequently, the electrical field emitted by the second signal via **110**B may overlap with the electrical field emitted by the first signal via **110**A to reduce an integrity of the respective signals propagated along the first signal via **110**A and along the second signal via **110**B. The reduced integrity of the signals may reduce operation of the PCB **100**.

[0020] For this reason, the PCB **100** may include components to reduce electrical field interference between the signal vias **110** to maintain desirable integrity of signals. As an example, respective ground vias may surround the signal vias **110**, and the respective ground vias may share a common ground via positioned between the vias **110**. Such an arrangement of the ground vias may block the electrical fields from interfering with one another, thereby improving signal integrity and operation of the PCB **100**.

[0021] FIG. **2** is a schematic diagram of a PCB **150**. The PCB **150** may include a first layer **152** (e.g., a top layer), a second layer **154** (e.g., a bottom layer), a third layer **156** (e.g., a first inner layer), and a fourth layer **158** (e.g., a second inner layer). Other layers of the PCB **150**, such as ground layers between the depicted layers **152**, **154**, **156**, **158**, are not illustrated for visualization purposes. The PCB **150** may also include a first pair of signal vias **160**, a second pair of signal vias **162**, a third pair of signal vias **164**, and a fourth pair of signal vias **166**. Each of the first pair of signal vias **160** and the second pair of signal vias **162** may extend from the first layer **152** to the third layer **156**, and each of the third pair of signal vias **164** and the fourth pair of signal vias **166** may extend from the second layer **154** to the fourth layer **158**. By way of example, the first pair of signal vias **160** and the second pair of signal vias **162** may each transmit respective signals from the first layer **152** toward the third layer **156**, and the third pair of signal vias **164** and the fourth pair of signal vias **166** may transmit respective signals from the second layer **154** toward the fourth layer **158**. The respective signals transmitted by each pair of signal vias **160**, **162**, **164**, **166** may include a positive signal and a negative signal in some embodiments.

[0022] The first pair of signal vias **160** and the second pair of signal vias **162** may overlap with the third pair of signal vias **164** and/or the fourth pair of signal vias **166** along an axis **168** (e.g., a vertical axis, a Z-axis). For instance, the fourth layer **158** at which the third pair of signal vias **164** and the fourth pair of signal vias **166** terminate may be adjacent to the first layer **152** (e.g., separated by relatively fewer ground layers) from which the first pair of signal vias **160** and the second pair of signal vias **162** extend, whereas the third layer **156** at which the first pair of signal vias **160** and the second pair of signal vias **162** terminate may be adjacent to the second layer **154** (e.g., separated by relatively fewer ground layers) from which the third pair of signal vias **164** and the fourth pair of signal vias **166** extend. Consequently, signals propagated along the first pair of signal vias **160** and/or along the second pair of signal vias **162** from the first layer **152** toward the third layer **156** may cause electrical fields to be emitted from the first pair of signal vias **160** and/or from the second pair of signal vias **162** toward the third pair of signal vias **164** and/or toward the fourth pair of signal vias **166**. Additionally or alternatively, signals propagated along the third pair of signal vias **164** and/or along the fourth pair of signal vias **166** from the second layer **154** toward the fourth layer **158** may cause electrical fields to be emitted from the third pair of signal vias **164** and/or from the fourth pair of signal vias **166** toward the first pair of signal vias **160** and/or toward the second pair of signal vias **162**. However, the PCB **100** may include ground vias (not shown) positioned to block such electrical fields from interfering with one another. By way of example, the ground vias may extend through any of the layers **152**, **154**, **156**, **158** to contain and isolate the electrical fields. As such, even though the first pair of signal vias **160** and the second pair of signal vias **162** may overlap with the third pair of signal vias **164** and the fourth pair of signal vias **166** along the axis **168** to cause electrical fields to be emitted from one of the pairs of signal vias **160**, **162**, **164**, **166** toward another (e.g., an adjacent) pair of the signal vias **160**, **162**, **164**, **166**, overlap between the electrical fields may be limited to maintain desirable integrity of the signals propagated along the pairs of signal vias **160**, **162**, **164**, **166**.

[0023] The overlap of adjacent pairs of signal vias **160**, **162**, **164**, **166** along the axis **168** may enable more efficient usage of the PCB **150** to propagate signals. As an example, adjacent pairs of signal vias **160**, **162**, **164**, **166** that overlap along the axis **168** may be positioned more proximate to one another while achieving desirable signal integrity. As another example, layers of the PCB **150** may be used more readily. For instance, instead of limiting extension of the pairs of signal vias **160**, **162**, **164**, **166** through the layers **152**, **154**, **156**, **158** to avoid overlap with one another along the axis **168** (e.g., by terminating the third pair of signal vias **164** and/or the fourth pair of signal vias **166** prior to the fourth layer **158** and/or by terminating the first pair of signal vias **160** and/or the second pair of signal vias **162** prior to the third layer **156**), each pair of signal via **160**, **162**, **164**, **166** may extend to and terminate at any of the layers **152**, **154**, **156**, **158**, such as to overlap with one another along the axis **168**, without compromising signal integrity. Additionally or alternatively, the PCB **150** may include fewer layers that otherwise may be used to offset the pairs of signal vias **160**, **162**, **164**, **166** along the axis **168**. In either case, the PCB **150** may be more efficiently utilized, such as by increasing a quantity of components that can be arranged on the PCB **150** and/or by reducing an amount of material used to manufacture the PCB **150** (e.g., an excessive quantity of layers of the PCB **150**). As a result, operation of the PCB **150** may be improved and/or a cost of manufacture of the PCB **150** may be reduced.

[0024] FIG. **3** is a top perspective view of the PCB **150** that includes the first pair of signal vias **160** and the second pair of signal vias **162** extending from the first layer **152** to the third layer **156** and the third pair of signal vias **164** and the fourth pair of signal vias **166** extending from the second layer **154** to the fourth layer **158**. The PCB **150** may also include first traces **200** routed along the first layer **152** and coupled to the first pair of signal vias **160**, as well as second traces **202** routed along the first layer **152** and coupled to the second pair of signal vias **162**. For example, the first traces **200** and the second traces **202** may propagate signals between the PCB **150** and a separate electrical component (e.g., an IC) at the first layer **152**. To this end, first interconnects **204** may be coupled to each of the first traces **200** and second traces **202** for coupling to the separate electrical component. Furthermore, third traces **206** may be routed along the third layer **156** and coupled to the first pair of signal vias **160**, and fourth traces **208** may be routed along the third layer **156** and coupled to the second pair of signal vias **162**, such as for connection with an electrical component at the third layer **156**. Thus, signals may be transmitted between the separate electrical component at the first layer **152** and the electrical component at the third layer **156** by way of the first interconnects **204**, the first traces **200**, the first pair of signal vias **160**, and the third traces **206** and/or by way of the first interconnects **204**, the second traces **202**, the second pair of signal vias **162**, and the fourth traces **208**.

[0025] The PCB **150** may further include fifth traces **210** routed along the second layer **154** and coupled to the third pair of signal vias **164**, as well as sixth traces **212** routed along the second layer **154** and coupled to the fourth pair of signal vias **166**. For instance, the fifth traces **210** and the sixth traces **212** may propagate signal between the PCB **150** and another separate component (e.g., another IC) at the second layer **154** using second interconnects **214** coupled to each of the fifth traces **210** and sixth traces **212**. As such, the PCB **150** may have a belly-to-belly configuration in which the first layer **152** and the second layer **154** at opposite sides of the PCB **150** are connected to a respective, separate electrical component. Moreover, seventh traces **216** may be routed along the fourth layer **158** and coupled to the third pair of signal vias **164**, and eighth traces **218** may be routed along the fourth layer **158** and coupled to the fourth pair of signal vias **166**, such as for connection with an electrical component at the fourth layer **158**. As such, signals may be transmitted between the separate electrical component at the second layer **154** and the electrical component at the fourth layer **158** by way of the second interconnects **214**, the fifth traces **210**, the third pair of signal vias **164**, and the seventh traces **216** and/or by way of the second interconnects **214**, the sixth traces **212**, the fourth pair of signal vias **166**, and the eighth trace **218**.

[0026] FIG. **4** is a schematic diagram of a layer **250** (e.g., an external layer) of the PCB **150**, such as of the first layer **152** and/or of the second layer **154**. A first pair of signal vias **252**, a second pair of signal vias **254**, and a third pair of signal vias **256** may be formed at the layer **250**. For instance, each pair of signal vias **252**, **254**, **256** may extend through the layer **250** along a first axis **258** in overlap with one another. First traces **260** may be coupled to the first pair of signal vias **252** and routed along the layer **250**, and second traces **262** may be coupled to the third pair of signal vias **256** and routed along the layer **250**, such as to couple the first pair of signal vias **252** and the third pair of signal vias **256** to other electrical components at the layer **250**. However, no traces routed along the layer **250** may be coupled to the second pair of signal vias **254**. Thus, the second pair of signal vias **254** may not be coupled to another electrical component at the layer **250**.

[0027] Signal propagation along the pairs of signal vias **252**, **254**, **256** may cause electrical fields to be emitted. To block interference between the electrical fields, ground vias may be arranged to isolate the electrical fields. By way of example, a first plurality of ground vias **264** may at least partially circumferentially surround a first signal via **266** of the first pair of signal vias **252**, a second plurality of ground vias **268** may at least partially circumferentially surround a second signal via **270** of the second pair of signal vias **254**, and the first plurality of ground vias **264** and the second plurality of ground vias **268** may share a first common ground via **272** positioned between the first signal via **266** and the second signal via **270**. That is, the first signal via **266** may be positioned at the same first distance **274** away from each of the first plurality of ground vias **264**, including the first common ground via **272**, and the second signal via **270** may be positioned at the same second distance **276** away from each of the second plurality of ground vias **268**, including the first common ground via **272**. For instance, the first distance **274** and the second distance **276** may be approximately equal to one another. Such an arrangement of the first plurality of ground vias **264** about the first signal via **266** and of the second plurality of ground vias **268** about the second signal via **270** may block electrical fields emitted by the first signal via **266** and electrical fields emitted by the second signal via **270** from interfering with one another. That is, the first plurality of ground vias **264** may isolate the electrical fields emitted by the first signal via **266**, and the second plurality of ground vias **268** may isolate the electrical fields emitted by the second signal via **270**. The first common ground via **272** may help isolate the electrical fields emitted by both the first signal via **266** and the second signal via **270**. As such, the first plurality of ground vias

264 and the second plurality of ground vias **268** may efficiently isolate the electrical fields to maintain desirable signal integrity (e.g., by using a limited quantity of ground vias and/or by enabling the first signal via **266** and the second signal via **270** to be positioned more adjacent to one another).

[0028] A third plurality of ground vias **278** may at least partially circumferentially surround a third signal via **280** of the second pair of signal vias **254**, a fourth plurality of ground vias **282** may at least partially circumferentially surround a fourth signal via **284** of the third pair of signal vias **256**, and the third plurality of ground vias **278** and the fourth plurality of ground vias **282** may share a second common ground via **286** positioned between the third signal via **280** and the fourth signal via **284**. As such, the third signal via **280** may be positioned at the same third distance **288** away from each of the third plurality of ground vias **278**, including the second common ground via **286**, and the fourth signal via **284** may be positioned at the same fourth distance **290** away from each of the fourth plurality of ground vias **282**, including the second common ground via **286**. For example, the third distance **288** and the fourth distance **290** may be approximately equal to one another. Thus, the third plurality of ground vias **278** and the fourth plurality of ground vias **282** may block electrical fields emitted by the third signal via **280** and electrical fields emitted by the fourth signal via **284** from interfering with one another to efficiently isolate the electrical fields and maintain desirable signal integrity. In some embodiments, the second distance **276** and the third distance **288** are approximately equal to one another such that the second plurality of ground vias **268** and the third plurality of ground vias **278** are symmetrical to one another about the second pair of signal vias **254**.

[0029] Ground vias may also be similarly positioned about a fifth signal via **292** of the first pair of signal vias **252** and/or about a sixth signal via **294** of the third pair of signal vias **256**. That is, a fifth plurality of ground vias **296** may at least partially circumferentially surround the fifth signal via **292** and/or a sixth plurality of ground vias **298** may at least partially circumferentially surround the sixth signal via **294**. For example, the fifth signal via **292** may be positioned at the first distance **274** away from each of the fifth plurality of ground vias **296**, and/or the sixth signal via **294** may be positioned at the fourth distance **290** away from each of the sixth plurality of ground vias **298**. Therefore, the first plurality of ground vias **264** and the fifth plurality of ground vias **296** may be symmetrical to one another about the first pair of signal vias **252** and/or the fourth plurality of ground vias **282** and the sixth plurality of ground vias **298** may be symmetrical to one another about the third pair of signal vias **256**.

[0030] In certain embodiments, each signal via **266**, **270**, **280**, **284**, **292**, **294** may be offset from one another along a second axis **300** (e.g., a first horizontal axis), perpendicular to the first axis **258**, and aligned with one another along a third axis **302** (e.g., a second horizontal axis), perpendicular to the first axis **258** and to the second axis **300**. The first common ground via **272** and the second common ground via **286** may also be offset from the signal vias **266**, **270**, **280**, **284**, **292**, **294** along the second axis **300** and aligned with the signal vias **266**, **270**, **280**, **284**, **292**, **294** along the third axis **302**. Thus, the pairs of signal vias **252**, **254**, **256** and the common ground vias **272**, **286** may be collinear with one another. Moreover, a first intermediate ground via **304** may be positioned between the first signal via **266** and the second signal via **270** along the second axis **300** and further help isolate electrical fields emitted by both the first signal via **266** and by the second signal via **270**. For instance, the first intermediate ground via **304** may be aligned with the first common ground via **272** along the second axis **300** and offset from the first common ground via **272** along the third axis **302**. The first intermediate ground via **304** may be positioned at a fifth distance **308**, greater than the first distance **274** and the second distance **276**, away from the first signal via **266** and away from the second signal via **270**. Therefore, the first plurality of ground vias **264**, the second plurality of ground vias **268**, and the first intermediate ground via **304** may cooperatively form a Y-shaped arrangement. A second intermediate ground via **306** may be positioned between the third signal via **280** and the fourth signal via **284** along the second axis **300** and further help isolate electrical fields emitted by both the third signal via **280** and by the fourth signal via **284**. The second intermediate ground via **306** may be aligned with the second common ground via **286** along the second axis **300** and offset from the second common ground via **286** along the third axis **302**. The second intermediate ground via **306** may be positioned at a sixth distance **310** (e.g., the same as the fifth distance **308**), greater than the third distance **288** and the fourth distance **290**, away from the third signal via **280** and away from the fourth signal via **284**. In this manner, the third plurality of ground vias **278**, the fourth plurality of ground vias **282**, and the second intermediate ground via **306** may cooperatively form a Y-shaped arrangement. Further still, additional, respective vias **312** may be positioned to isolate electric fields emitted by the fifth signal via **292** and by the sixth signal via **294**.

[0031] The first plurality of ground vias **264** and the fifth plurality of ground vias **296** may provide sufficient space for the first traces **260** to be routed from the first pair of signal vias **252** in directions away from the first plurality of ground vias **264** and from the fifth plurality ground vias **296**. In addition, the fourth plurality of ground vias **282** and the sixth plurality of ground vias **298** may provide sufficient space for the second traces **262** to be routed from the third pair of signal vias **256** in directions away from the fourth plurality of ground vias **282** and from the sixth plurality of ground vias **298**. Respective ground microvias **314** may surround the traces **260**, **262** to block interference of electrical fields emitted as a result of signal propagation along the traces **260**, **262**. Thus, the ground microvias **314** may further help maintain signal integrity. Although the traces **260**, **262** may transmit signals along the layer **250** in the illustrated embodiment, another component, such as a microstrip and/or an interconnect, may be used for transmitting signals along the layer **250** in additional or alternative embodiments.

[0032] Additionally, in the illustrated embodiment, no ground vias are positioned between the first plurality of ground vias **264** and the fifth plurality of ground vias **296** along the second axis **300**, between the second plurality of ground vias **268** and the third plurality of ground vias **278** along the second axis **300**, and between the fourth plurality of ground vias **282** and the sixth plurality of ground vias **298** along the second axis **300**. Accordingly, six respective ground vias surround each pair of signal vias **252**, **254**, **256** to sufficiently isolate electrical fields emitted by the pairs of signal vias **252**, **254**, **256**.

[0033] In some embodiments, each of the ground vias **264**, **268**, **278**, **282**, **296**, **298**, **304**, **306**, **312** may extend through multiple layers of the PCB **150**. For example, each of the signal vias **266**, **270**, **280**, **284**, **292**, **294** may extend through another layer of the PCB **150** in addition to the layer **250**, and the ground vias **264**, **268**, **278**, **282**, **296**, **298**, **304**, **306**, **312** may extend through corresponding layers to isolate electrical fields emitted by the signal vias **266**, **270**, **280**, **284**, **292**, **294** at each layer. However, the traces **260**, **262** may be routed along the layer **250** and not along another layer of the PCB **150**. Therefore, the ground microvias **314** may extend through the layer **250** and not through another layer of the PCB **150** to sufficiently isolate electrical fields emitted by the traces **260**, **262**.

[0034] FIG. **5** is a schematic diagram of a layer **350** (e.g., an internal layer) of the PCB **150**, such as of the third layer **156** and/or of the fourth layer **158**. A first pair of signal vias **352** and a second pair of signal vias **354** may be formed at the layer **350**. A first plurality of ground vias **356** may at least partially circumferentially surround a first signal via **358** of the first pair of signal vias **352**, a second plurality of ground vias **360** may at least partially circumferentially surround a second signal via **362** of the first pair of signal vias **352**, a third plurality of ground vias **364** may at least partially circumferentially surround a third signal via **366** of the second pair of signal vias **354**, and a fourth plurality of ground vias **368** may at least partially circumferentially surround a fourth signal via **370** of the second pair of signal vias **354**. In the FIG. **5**, no traces are illustrated. However, in some embodiments, traces may be coupled to the first pair of signal vias **352** and/or the second pair of signal vias **354**. For example, the traces routed along the layer **350** may be relatively narrower as compared to the traces **260**, **262** routed along the layer **250**. As such, the traces may be accommodated by and routed through a smaller space. For this reason, a greater quantity of ground vias **356**, **360**, **364**, **368** may more fully surround each signal via **358**, **362**, **366**, **370**, respectively, as compared to the respective ground vias surrounding the signal vias **266**, **284**, **292**, **294** to which traces **260**, **262** are coupled at the layer **250**, while providing a sufficient amount of space to enable traces to be routed between the ground vias **356**, **360**, **364**, **368**. For instance, ten respective ground vias may surround each pair of signal vias **352**, **354**.

[0035] By way of example, the second plurality of ground vias **360** and the third plurality of ground vias **364** may share a common ground via **372**. In some embodiments, the signal vias **358**, **362**, **366**, **370** and the common ground via **372** may be offset from one another along a first axis **374** (e.g., a first horizontal axis) and aligned with one another along a second axis **376** (e.g., a second horizontal axis). The second plurality of ground vias **360** may also include a first ground via **378** and a second ground via **380**. The first ground via **378** and the second ground via **380** may be aligned with one another along the first axis **374** and offset from one another along the second axis **376**. The third plurality of ground vias **364** may include a third ground via **382** and a fourth ground via **384** that are aligned with one another along the first axis **374** and offset from one another along the second axis **376**. As an example, the third ground via **382** may be aligned with the first ground via **378** along the second axis **376**, and the fourth ground via **384** may be aligned with the second ground via **380** along the second axis **376**. A first intermediate ground via **386** may be positioned adjacent to the first ground via **378** and to the third ground via **382**, offset from the common ground via **372** along the second axis **376**, and aligned with the common ground via **372** along the first axis **374**. A second intermediate ground via **388** may be positioned adjacent to the second ground via **380** and to the fourth ground via **384**, offset from the common ground via **372** along the second axis **376**, and aligned with the common ground via **372** along the first axis **374**. As such, the second plurality of ground vias **360**, the third plurality of ground vias **364**, the first intermediate ground via **386**, and the second intermediate ground via **388** may cooperatively isolate electrical fields emitted from the second signal via **362** and from the third signal via **366**.

[0036] The first plurality of ground vias **356** may include a fifth ground via **390** aligned with the first signal via **358** along the second axis **376** and offset from the first signal via **358** along the first axis **374**, as well as a sixth ground via **392** and a seventh ground via **394** aligned with one another along the first axis **374** and offset from one another along the second axis **376**. For example, the sixth ground via **392** may be aligned with the first ground via **378** along the second axis **376**, and the seventh ground via **394** may be aligned with the second ground via **380** along the second axis **376**. A third intermediate ground via **396** positioned adjacent to the sixth ground via **392** and a fourth intermediate ground via **398** positioned adjacent to the seventh ground via **394** may be aligned with one another along the first axis **374** and offset from one another along the second axis **376**. For instance, the third intermediate ground via **396** may be aligned with the first intermediate ground via **386** along the second axis **376**, and the fourth intermediate ground via **398** may be aligned with the second intermediate ground via **388** along the second axis **376**. In this manner, the first plurality of ground vias **356**, the second plurality of ground vias **360**, and the intermediate ground vias **386**, **388**, **396**, **398** may be symmetrical to one another about the first pair of signal vias **352**.

[0037] Similarly, the fourth plurality of ground vias **368** may include an eighth ground via **400** aligned with the fourth signal via **370** along the second axis **376** and offset from the fourth signal via **370** along the first axis **374**, as well as a ninth ground via **402** and a tenth ground via **404** aligned with one another along the first axis **374** and offset from one another along the second axis **376**. The ninth ground via **402** may be aligned with the third ground via **382** along the second axis **376**, and the tenth ground via **404** may be aligned with the fourth ground via **384** along the second axis **376**. A fifth intermediate ground via **406** positioned adjacent to the ninth ground via **402** and a sixth intermediate ground via **408** positioned adjacent to the tenth ground via **404** may be aligned with one another along the first axis **374** and offset from one another along the second axis **376**. As an example, the fifth intermediate ground via **406** may be aligned with the first intermediate ground via **386** along the second axis **376**, and the sixth intermediate ground via **408** may be aligned with the second intermediate ground via **388** along the second axis **376**. As such, the third plurality of ground vias **364**, the fourth plurality of ground vias **368**, and the intermediate ground vias **386**, **388**, **406**, **408** may be symmetrical to one another about the second pair of signal vias **354**. Indeed, the ground vias **378**, **380**, **390**, **392**, **394**, **396**, **398** cooperatively surrounding the first pair of signal vias **352** and the ground vias **382**, **384**, **400**, **402**, **404**, **406**, **408** cooperatively surrounding the second pair of signal vias **354** may be symmetrical to one another about the ground vias **372**, **386**, **388** positioned between the pairs of signal vias **352**, **354**.

[0038] In some embodiments, the respective center of each of the first plurality of ground vias **356** may be at a first distance **410** away from the center of the first signal via **358**, and the respective center of each of the second plurality of ground vias **360** may also be at the first distance **410** away from the center of the second signal via **362**. Similarly, the respective center of each of the third plurality of ground vias **364** may be at the first distance **410** away from the center of the third signal via **366**, and the respective center of each of the fourth plurality of ground vias **368** may be at the first distance **410** away from the center of the fourth signal via **370**. By way of example, the first distance **410** may be a value between 0.5 millimeters (mm) and 0.7 mm (i.e., between 0.02 inches (in) and 0.028 in). Additionally, the center of the first signal via **358** may be at a second distance **412** away from the center of the second signal via **362**. Similarly, the center of the third signal via **366** may be at the second distance **412** away from the center of the fourth signal via **370**. For example, the second distance **412** may be substantially similar to the first distance **410**. The respective centers of the first intermediate via **386** and of the second intermediate via **388** may be at a third distance **414** away from the center of the common ground via **372**. The third distance **414** may be substantially less than the first distance **410** and the second distance **412**. For instance, the third distance **414** may be a value between 0.35 mm and 0.4 mm (i.e., between 0.014 in and 0.016 in). The respective centers of the third intermediate via **396** and of the fourth intermediate via **398** may also be at the third distance **414** away from the center of the fifth ground via **390**, and/or the respective centers of the fifth intermediate ground via **406** and of the sixth intermediate ground via **408** may be at the third distance **414** away from the center of the eighth ground via **400**. Further still, the center of the first ground via **378** may be at a fourth distance **416** away from the center of the sixth ground via **392**, the center of the second ground via **380** may be at the fourth distance **416** away from the center of the seventh ground via **394**, the center of the third ground via **382** may be at the fourth distance **416** away from the center of the ninth ground via **402**, and/or the center of the fourth ground via **384** may be at the fourth distance **416** away from the center of the tenth ground via **404**. The fourth distance **416** may be substantially greater than each of the first distance **410**, the second distance **412**, and the third distance **414**. As an example, the fourth distance **416** may be a value between 0.84 mm and 0.91 mm (i.e., between 0.033 in and 0.036 in).

[0039] FIG. **6** is a schematic diagram of a layer **500** of the PCB **150**, such as of the first layer **152**, of the second layer **154**, of the third layer **156**, and/or of the fourth layer **158**. The layer **500** may be a power layer configured to receive and direct power to electrical components at the layer **500**. However, the layer **500** may also include a ground portion **502** that does not receive and direct power. For example, the layer **500** may include a power section **504** (e.g., a power shape, a power plane) with a cutout or opening **506**, and the ground portion **502** (e.g., a ground shape, a ground plane) may be positioned within the cutout **506** to avoid contact with the power section **504**. The ground portion **502** may therefore be isolated from the power directed by the power section **504**. As such, the ground portion **502** may also avoid redirecting power away from the power section **504**, thereby enabling power to be more efficiently utilized at the layer **500** by the power section **504**, such as by providing a sufficient current path along the power section **504** for power to travel.

[0040] Additionally, pairs of signal vias **508** may extend through the layer **500** at the ground portion **502** such that the ground portion **502** blocks power from being directed to the signal vias

508 at the layer **500**. Ground vias **510** may be positioned about the pairs of signal vias **508** in a similar arrangement as that described with respect to FIG. **5**. That is, respective ground vias **510** may at least partially circumferentially surround each of the signal vias **508**, and a respective common ground via **512** (e.g., a single common ground via **512**) may be positioned between adjacent pairs of signal vias **508**.

[0041] The ground portion **502** may help maintain desirable integrity of signals propagated along the signal vias **508** at the layer **500**. For example, the ground portion **502** may limit resonance that otherwise may reduce signal integrity and that otherwise may be present without the ground portion **502** (e.g., to isolate the signal vias **508** from the power section **504** by a large void provided with the cutout **506**). However, in additional or alternative embodiments, the layer **500** may not include the ground portion **502**, and the signal vias **508** and the ground vias **510** may therefore extend through the space provided by the cutout **506** to avoid receiving power.

[0042] Each of FIGS. **7** and **8** discussed below illustrates a respective method for manufacturing a PCB, such as any of the PCBs **100**, **150** discussed herein. In certain embodiments, the methods may be performed differently than depicted. For example, an additional operation may be performed, any of the operations may be performed differently than depicted, any of the operations may not be performed, and/or the operations may be performed in a different order. Furthermore, the respective operations of the methods may be performed in any manner relative to one another, such as sequentially and/or concurrently.

[0043] FIG. **7** is a flowchart of a method **550** for manufacturing a PCB. At step **552**, adjacent pairs of signal vias may be formed through a layer of the PCB. The signal vias of the pairs of signal vias may be aligned with one another along a first horizontal axis and offset from one another along a second horizontal axis, perpendicular to the first horizontal axis. The signal vias may be configured to propagate respective signals. For example, each pair of signal vias may be a differential pair in which one of the signal vias transmits a positive signal and the other of the signal vias transmits a corresponding negative signal.

[0044] At step **554**, a ground via assembly is formed around the adjacent pairs of signal vias through the layer of the PCB. The ground via assembly may include a first plurality of ground vias that at least partially circumferentially surrounds a first signal via of a first pair of signal vias and a second plurality of ground vias that at least partially circumferentially surrounds a second signal via of a second pair of signal vias. The first plurality of ground vias may isolate electrical fields emitted by the first signal via (e.g., toward the second signal via), and the second plurality of ground vias may isolate electrical fields emitted by the second signal via (e.g., toward the first signal via). Thus, the first plurality of ground vias and the second plurality of ground vias may cooperatively block respective electrical fields emitted by the first signal via and by the second signal via from interfering with one another, thereby helping maintain integrity of respective signals transmitted by the first signal via and by the second signal via.

[0045] Additionally, the first plurality of ground vias and the second plurality of ground vias may share a common ground via positioned between the first signal via and the second signal via. For instance, the common ground via may be aligned with the first signal via and with the second signal via along the first horizontal axis and offset from the first signal via and from the second signal via along the second horizontal axis. In other words, the pairs of signal vias may be offset from one another along the second horizontal axis by a single ground via (i.e., the common ground via). The common ground via may help isolate each of the respective electrical fields emitted by the first signal via and by the second signal via. As a result, the ground via assembly may enable the pairs of signal vias to be positioned more adjacent to one another without reducing signal integrity.

[0046] At step **556**, traces may be coupled to any of the pairs of signal vias at the layer of the PCB. To this end, the ground via assembly (e.g., the first plurality of ground vias, the second plurality of ground vias) may be positioned to provide a sufficient amount of space to enable routing of the traces from the signal vias along the layer. By way of example, traces coupled to the first pair of signal vias may extend away from the first plurality of ground vias, and traces coupled to the second pair of signal vias may extend away from the second plurality of ground vias. The traces may enable signal transmission from the signal vias along the layer of the PCB, such as to another electrical component at the layer.

[0047] At step **558**, ground microvias may be formed around the traces at the layer. The ground microvias may block electrical fields emitted from the traces as a result of signal propagation along the traces. Thus, the ground microvias may further help maintain desirable signal integrity. In some embodiments, the ground vias of the ground via assembly may extend through each layer through which the pairs of signal vias extend to isolate electrical fields emitted by the signal vias at each layer. However, the ground microvias may extend through the layer and not through other layers of the PCB to isolate electrical fields emitted by the traces, which may be routed along the layer and not along other layers of the PCB.

[0048] FIG. **8** is a flowchart of an embodiment of a method **600** for manufacturing a PCB. At step **602**, adjacent pairs of signal vias may be formed through a power layer of the PCB. For instance, the power layer may include a power portion (e.g., a power shape, a power plane) configured to direct power therethrough. The power portion may include a cutout or opening, and the pairs of signal vias may be positioned in the cutout to avoid contact with the power portion. As such, the pairs of signal vias may avoid receiving power directed through the power portion.

[0049] At step **604**, a ground portion (e.g., a ground shape, a ground plane) may be added to the power layer around the adjacent pairs of signal vias, such as by positioning the ground portion within the cutout of the power portion. The ground portion may isolate the pairs of signal vias from the power portion while reducing or limiting resonance that otherwise may be caused by an abundance of space surrounding the signal vias (e.g., as provided by the cutout). Thus, the ground portion may help improve signal integrity.

[0050] At step **606**, a ground via assembly is formed through the ground portion to surround the adjacent pairs of signal vias. The ground via assembly may include a first plurality of ground vias that at least partially circumferentially surrounds a first signal via of a first pair of signal vias and a second plurality of ground vias that at least partially circumferentially surrounds a second signal via of a second pair of signal vias to isolate respective electrical fields emitted by the first signal via and by the second signal via. The first plurality of ground vias and the second plurality of ground vias may share a common ground via that may help isolate each respective electrical field emitted by the first signal via and by the second signal via. Thus, the ground via assembly may enable the pairs of signal vias to be positioned more adjacent to one another without reducing signal integrity.

[0051] In some aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; a first pair of signal vias extending through the layer and configured to propagate respective signals; a second pair of signal vias extending through the layer and configured to propagate respective signals; a first plurality of ground vias extending through the layer and at least partially circumferentially surrounding a first signal via of the first pair of signal vias; and a second plurality of ground vias extending through the layer and at least partially circumferentially surrounding a second signal via of the second pair of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias share a common ground via.

[0052] In some aspects, the techniques described herein relate to an apparatus, wherein the first pair of signal vias, the second pair of signal vias, and the common ground via are offset from one another along a first axis and aligned with one another along a second axis that is perpendicular to the first axis.

[0053] In some aspects, the techniques described herein relate to an apparatus, including an additional ground via extending through the layer, wherein the additional ground via is aligned with the common ground via along the first axis and offset from the common ground via along the second axis.

[0054] In some aspects, the techniques described herein relate to an apparatus, wherein the first plurality of ground vias includes a first ground via and a second ground via, wherein the first ground via and the second ground via are aligned with one another along the first axis and offset from one another along the second axis.
[0055] In some aspects, the techniques described herein relate to an apparatus, wherein the first plurality of ground vias includes a first ground via, the second plurality of ground vias includes a second ground via, and the first ground via and the second ground via are offset from one another along the first axis and aligned with one another along the second axis.
[0056] In some aspects, the techniques described herein relate to an apparatus, including a third plurality of ground vias extending through the layer and at least partially surrounding a third signal via of the first pair of signal vias, wherein the first plurality of ground vias includes a first ground via, the third plurality of ground vias includes a second ground via, and the first ground via and the second ground via are offset from one another along the first axis and aligned with one another along the second axis.
[0057] In some aspects, the techniques described herein relate to an apparatus, including a trace coupled to the first signal via or the third signal via of the first pair of signal vias, wherein the trace is routed along the layer of the printed circuit board in a direction away from the first ground via or away from the second ground via.
[0058] In some aspects, the techniques described herein relate to an apparatus, including a plurality of ground microvias formed through the layer and surrounding the trace.
[0059] In some aspects, the techniques described herein relate to an apparatus, wherein the layer is a first outer layer extending along a first side of the printed circuit board, the printed circuit board includes a second outer layer extending along a second side, opposite the first side, of the printed circuit board, the apparatus includes a third pair of signal vias extending through the second outer layer and configured to propagate respective signals, and the third pair of signal vias extends in overlap with the first pair of signal vias and/or with the second pair of signal vias at an inner layer between the first outer layer and the second outer layer.
[0060] In some aspects, the techniques described herein relate to an apparatus, wherein the layer is a power layer that includes a ground portion, and the first plurality of ground vias and the second plurality of ground vias extend through the ground portion.
[0061] In some aspects, the techniques described herein relate to a method including: forming adjacent pairs of signal vias through a layer of a printed circuit board; forming a first plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a first signal via of a first pair of signal vias of the adjacent pairs of signal vias; and forming a second plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a second signal via of a second pair of signal vias of the adjacent pairs of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias include a common ground via.
[0062] In some aspects, the techniques described herein relate to a method, wherein the layer includes a power shape with a cutout, and the adjacent pairs of signal vias, the first plurality of ground vias, and the second plurality of ground vias are formed through the cutout.
[0063] In some aspects, the techniques described herein relate to a method, further including positioning a ground shape within the cutout, wherein the adjacent pairs of signal vias, the first plurality of ground vias, and the second plurality of ground vias are formed through ground shape.
[0064] In some aspects, the techniques described herein relate to a method, further including coupling a trace to the first signal via or to the second signal via and routing the trace along the layer.
[0065] In some aspects, the techniques described herein relate to a method, wherein the layer is a first outer layer extending along a first side of the printed circuit board, the printed circuit board includes a second outer layer extending along a second side, opposite the first side, of the printed

circuit board, and the method further includes forming an additional pair of signal vias extending through the second outer layer in overlap with the adjacent pair of signal vias at an inner layer between the first outer layer and the second outer layer.

[0066] In some aspects, the techniques described herein relate to an apparatus including: a first pair of signal vias of a printed circuit board, wherein the first pair of signal vias are configured to propagate respective signals; a second pair of signal vias of the printed circuit board, wherein the second pair of signal vias are configured to propagate respective signals, and the signal vias of the first pair of signal vias and the second pair of signal vias are offset from one another along a first axis and aligned with one another along a second axis that is perpendicular to the first axis; and a single ground via positioned between the first pair of signal vias and the second pair of signal vias along the first axis.

[0067] In some aspects, the techniques described herein relate to an apparatus, wherein the single ground via is positioned equidistant to a first signal via of the first pair of signal vias and a second signal via of the second pair of signal vias.

[0068] In some aspects, the techniques described herein relate to an apparatus, including: a first plurality of ground vias circumferentially surrounding the first signal via of the first pair of signal vias; and a second plurality of ground vias circumferentially surrounding the second signal via of the second pair of signal vias, wherein the single ground via is of both the first plurality of ground vias and the second plurality of ground vias.

[0069] In some aspects, the techniques described herein relate to an apparatus, wherein a first signal via of the first pair of signal vias is positioned equidistant to the single ground via and a second signal via of the first pair of signal vias.

[0070] In some aspects, the techniques described herein relate to an apparatus, wherein the first pair of signal vias, the second pair of signal vias, and the single ground via extend through a cutout of a power shape of the printed circuit board to avoid contact with the power shape.

[0071] The above description is intended by way of example only. Although the techniques are illustrated and described herein as embodied in one or more specific examples, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made within the scope and range of equivalents of the claims.

[0072] As used herein, unless expressly stated to the contrary, use of the phrase ‘at least one of’, ‘one or more of’, ‘and/or’, variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions ‘at least one of X, Y and Z’, ‘at least one of X, Y or Z’, ‘one or more of X, Y and Z’, ‘one or more of X, Y or Z’ and ‘X, Y and/or Z’ can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.

[0073] Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in ‘one embodiment’, ‘example embodiment’, ‘an embodiment’, ‘another embodiment’, ‘certain embodiments’, ‘some embodiments’, ‘various embodiments’, ‘other embodiments’, ‘alternative embodiment’, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments.

[0074] Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

[0075] Additionally, unless expressly stated to the contrary, the terms ‘first’, ‘second’, ‘third’, etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node,

module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, ‘first X’ and ‘second X’ are intended to designate two ‘X’ elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, ‘at least one of’ and ‘one or more of’ can be represented using the ‘(s)’ nomenclature (e.g., one or more element(s)).
[0076] As used herein, the terms “approximately,” “generally,” “substantially,” and so forth, are intended to convey that the property value being described may be within a relatively small range of the property value, as those of ordinary skill would understand. For example, when a property value is described as being “approximately” equal to (or, for example, “substantially similar” to) a given value, this is intended to convey that the property value may be within +/−5%, within +/−4%, within +/−3%, within +/−2%, within +/−1%, or even closer, of the given value. Similarly, when a given feature is described as being “substantially parallel” to another feature, “generally perpendicular” to another feature, and so forth, this is intended to convey that the given feature is within +/−5%, within +/−4%, within +/−3%, within +/−2%, within +/−1%, or even closer, to having the described nature, such as being parallel to another feature, being perpendicular to another feature, and so forth. Mathematical terms, such as “parallel” and “perpendicular,” should not be rigidly interpreted in a strict mathematical sense, but should instead be interpreted as one of ordinary skill in the art would interpret such terms. For example, one of ordinary skill in the art would understand that two lines that are substantially parallel to each other are parallel to a substantial degree, but may have minor deviation from exactly parallel.
[0077] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible, or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).
[0078] One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

Claims

1. An apparatus comprising: a layer of a printed circuit board; a first pair of signal vias extending through the layer and configured to propagate respective signals; a second pair of signal vias extending through the layer and configured to propagate respective signals; a first plurality of ground vias extending through the layer and at least partially circumferentially surrounding a first signal via of the first pair of signal vias; and a second plurality of ground vias extending through the layer and at least partially circumferentially surrounding a second signal via of the second pair of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias share a common ground via.
2. The apparatus of claim 1, wherein the first pair of signal vias, the second pair of signal vias, and the common ground via are offset from one another along a first axis and aligned with one another along a second axis that is perpendicular to the first axis.
3. The apparatus of claim 2, further comprising an additional ground via extending through the

layer, wherein the additional ground via is aligned with the common ground via along the first axis and offset from the common ground via along the second axis.
4. The apparatus of claim 2, wherein the first plurality of ground vias comprises a first ground via and a second ground via, wherein the first ground via and the second ground via are aligned with one another along the first axis and offset from one another along the second axis.
5. The apparatus of claim 2, wherein the first plurality of ground vias comprises a first ground via, the second plurality of ground vias comprises a second ground via, and the first ground via and the second ground via are offset from one another along the first axis and aligned with one another along the second axis.
6. The apparatus of claim 2, further comprising a third plurality of ground vias extending through the layer and at least partially surrounding a third signal via of the first pair of signal vias, wherein the first plurality of ground vias comprises a first ground via, the third plurality of ground vias comprises a second ground via, and the first ground via and the second ground via are offset from one another along the first axis and aligned with one another along the second axis.
7. The apparatus of claim 6, further comprising a trace coupled to the first signal via or the third signal via of the first pair of signal vias, wherein the trace is routed along the layer of the printed circuit board in a direction away from the first ground via or away from the second ground via.
8. The apparatus of claim 7, further comprising a plurality of ground microvias formed through the layer and surrounding the trace.
9. The apparatus of claim 1, wherein the layer is a first outer layer extending along a first side of the printed circuit board, the printed circuit board comprises a second outer layer extending along a second side, opposite the first side, of the printed circuit board, the apparatus comprises a third pair of signal vias extending through the second outer layer and configured to propagate respective signals, and the third pair of signal vias extends in overlap with the first pair of signal vias and/or with the second pair of signal vias at an inner layer between the first outer layer and the second outer layer.
10. The apparatus of claim 1, wherein the layer is a power layer that comprises a ground portion, and the first plurality of ground vias and the second plurality of ground vias extend through the ground portion.
11. A method comprising: forming adjacent pairs of signal vias through a layer of a printed circuit board; forming a first plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a first signal via of a first pair of signal vias of the adjacent pairs of signal vias; and forming a second plurality of ground vias through the layer of the printed circuit board to at least partially circumferentially surround a second signal via of a second pair of signal vias of the adjacent pairs of signal vias, wherein the first plurality of ground vias and the second plurality of ground vias comprise a common ground via.
12. The method of claim 11, wherein the layer comprises a power shape with a cutout, and the adjacent pairs of signal vias, the first plurality of ground vias, and the second plurality of ground vias are formed through the cutout.
13. The method of claim 12, further comprising positioning a ground shape within the cutout, wherein the adjacent pairs of signal vias, the first plurality of ground vias, and the second plurality of ground vias are formed through ground shape.
14. The method of claim 11, further comprising coupling a trace to the first signal via or to the second signal via and routing the trace along the layer.
15. The method of claim 11, wherein the layer is a first outer layer extending along a first side of the printed circuit board, the printed circuit board comprises a second outer layer extending along a second side, opposite the first side, of the printed circuit board, and the method further comprises forming an additional pair of signal vias extending through the second outer layer in overlap with the adjacent pair of signal vias at an inner layer between the first outer layer and the second outer layer.

16. An apparatus comprising: a first pair of signal vias of a printed circuit board, wherein the first pair of signal vias are configured to propagate respective signals; a second pair of signal vias of the printed circuit board, wherein the second pair of signal vias are configured to propagate respective signals, and the signal vias of the first pair of signal vias and the second pair of signal vias are offset from one another along a first axis and aligned with one another along a second axis that is perpendicular to the first axis; and a single ground via positioned between the first pair of signal vias and the second pair of signal vias along the first axis.

17. The apparatus of claim 16, wherein the single ground via is positioned equidistant to a first signal via of the first pair of signal vias and a second signal via of the second pair of signal vias.

18. The apparatus of claim 17, further comprising: a first plurality of ground vias circumferentially surrounding the first signal via of the first pair of signal vias; and a second plurality of ground vias circumferentially surrounding the second signal via of the second pair of signal vias, wherein the single ground via is of both the first plurality of ground vias and the second plurality of ground vias.

19. The apparatus of claim 16, wherein a first signal via of the first pair of signal vias is positioned equidistant to the single ground via and a second signal via of the first pair of signal vias.

20. The apparatus of claim 16, wherein the first pair of signal vias, the second pair of signal vias, and the single ground via extend through a cutout of a power shape of the printed circuit board to avoid contact with the power shape.